

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3 –Poster Presentation	Assessment:	Assessment Tool 1– Poster Presentation
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	2023-B.Tech(AI&DS)	Date:	18-08-26

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	



Data Preparation and Statistical Analysis of Hospital Patient Records



Course Code : DSA0405

Course Name : Fundamentals of Data Science

Submitted by:

Jothika M(192324235) B.Tech(AI &DS)

Guided by:

Dr. Krishnaraj M

Introduction

Problem

How do we turn a messy, incomplete hospital EHR export into a dataset trustworthy enough to analyse?

Approach

Clean, encode & statistically analyse patient encounter records end-to-end

Why it matters

- Growing EHR adoption
- Need for reproducible clinical analytics
- Mature open-source tooling

Method Pipeline

Admissions Table

Field Standardisation

Vitals Table

Missing-Value Imputation

Labs Table

Outlier Capping (IQR)

Validation checks

- ✓ De-duplicate on encounter ID
- ✓ Admission / discharge date logic verified
- ✓ Cross-table join integrity checked

Analysis-Ready Dataset

Conclusion

1. Cleaning & encoding turned a noisy EHR extract into a consistent, analysis-ready dataset with 0% missingness.
2. Statistical tests confirm measurable, clinically plausible relationships between vitals, admission type and length of stay.
3. Outlier capping stabilised variance in length-of-stay without distorting its underlying distribution.

Future Directions

- o Extend the pipeline into a predictive model for 30-day readmission risk.
- o Automate the cleaning pipeline as a reusable script for new hospital extracts.
- o Validate findings against a second, independent hospital dataset.

Dataset

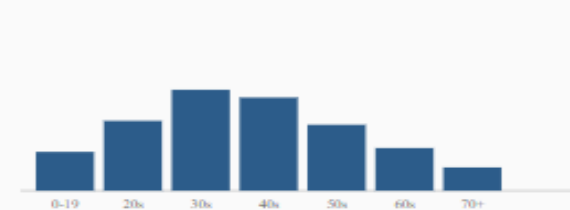
ID	AGE	BP	GLU	DIAG	LOS
1042	54	132	NA	I10	4
1043	67	NA	118	E11	7
1044	29	121	95	J45	2
1045	73	146	210	I50	15*

FIG. 1 — SAMPLE RAW RECORDS - ORANGE = MISSING / OUTLIER VALUES

- o De-identified hospital encounter extract used for coursework (teaching dataset).
- o 5,000 patient encounters across 18 variables: demographics, vitals, labs, diagnosis, length of stay (LOS), readmission flag.
- o Records span multiple admission types (emergency, elective, urgent) and diagnosis categories.
- o Split 80 / 20 into development and hold-out sets for downstream modelling.

Results

Age Distribution (n = 5,000)



Correlation Matrix

	BMI	GLU	BP
MI	1.0	.61	
LU	.61	1.0	
BP			1.0

Statistical Test Outcomes — Effect Size vs. Significance



BMI ↔ Glucose
r = 0.61

Readmit LOS t-test
p = 0.002

Glucose × Admission
ANOVA
p < 0.01

LOS variance after
capping
-18%